ACADEMIC PROJECT DOCUMENTATION

Student Enrollment & Course Management System

Phase 3: UI/UX Development & Customization

Platform: Salesforce Developer Edition (Lightning Experience)

PROJECT TEAM

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Note. Section 1 is the cover page. **Status key used in this document. Implemented** = shown in the supplied project screenshots. **Documented** = described in the project guide but no screenshot was supplied. **Planned** = not yet evidenced. **Requires confirmation** = information missing or inconsistent in the sources.

2. Phase 3 Overview

The Student Enrollment & Course Management System (SECMS) is a Salesforce application that centralises student, course, enrollment and instructor data. Earlier phases established the ideation, requirements, design and the underlying data model (four custom objects, their fields and lookup relationships). Phase 3 concentrates on the **user interface**: how enrollment officers, administrators and instructors see, enter and navigate the data.

The purpose of Phase 3 is to provide a user-friendly interface for managing students, courses, enrollments and instructors using standard Salesforce Lightning capabilities, without custom code.

Phase 3 Objectives

- Provide one Lightning app with clearly labelled tabs for each object.
- Arrange fields on page layouts so that only relevant information is shown and data entry is clear.
- Separate the first-time enrollment and re-enrollment experiences using record types and matching layouts.
- Give users helpful validation messages at the point of data entry.
- Display associated records through related lists and summarise activity on a dashboard.

Implementation Status Summary

Feature	Status	Evidence
Lightning app with navigation items	Implemented	Figures 2 and 3
Custom object tab (Students)	Implemented	Figure 1
Tabs for Courses, Instructors, Enrollments	Implemented (visible in app navigation)	Figure 3; separate tab-creation screenshots not supplied
Enrollment record types (New Enrollment, Re-Enrollment)	Implemented	Figures 4 and 5
Enrollment page layouts (New / Re-Enrollment)	Implemented	Figures 6 to 8
Page layout assignment by profile and record type	Documented	No screenshot; requires confirmation
Enrollments related list on Student, Course, Instructor layouts	Documented	No screenshot; requires confirmation
Validation messages (Student email, Fees paid)	Implemented	Figures 10 and 11
Dashboard for management view	Implemented	Figure 12
Layouts for Student, Course, Instructor objects	Requires confirmation	No screenshots supplied
Lightning record pages (App Builder)	Planned / requires confirmation	No evidence supplied
UI test execution	Not yet executed	Section 10

3. Salesforce User Interface

This section explains the Salesforce interface components used in the project.

3.1 App Launcher and Navigation

Users open the application from the App Launcher. The project provides a Lightning app named **Student Enrollment & Course Management** (developer name *Student_Enrollment_Course_Management*) with the primary colour #0070D2, as shown in Figure 2. The project guide selects Standard Navigation. The navigation bar contains the items selected in App Manager (Figure 3).

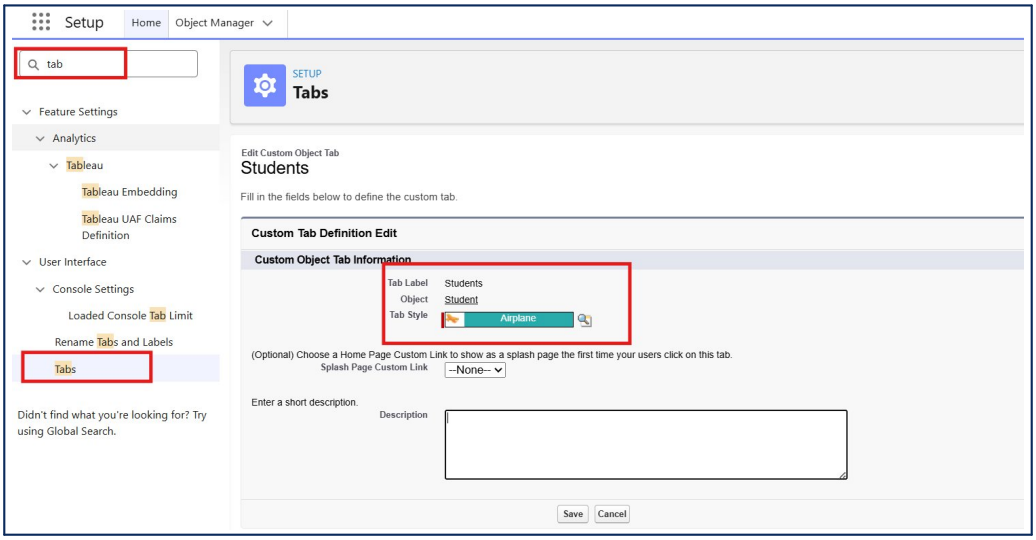


Figure 1. *The custom object tab for Students, created under Setup > Tabs. The Object field is set to Student and the tab label is Students. This makes the object available as a navigation item.*

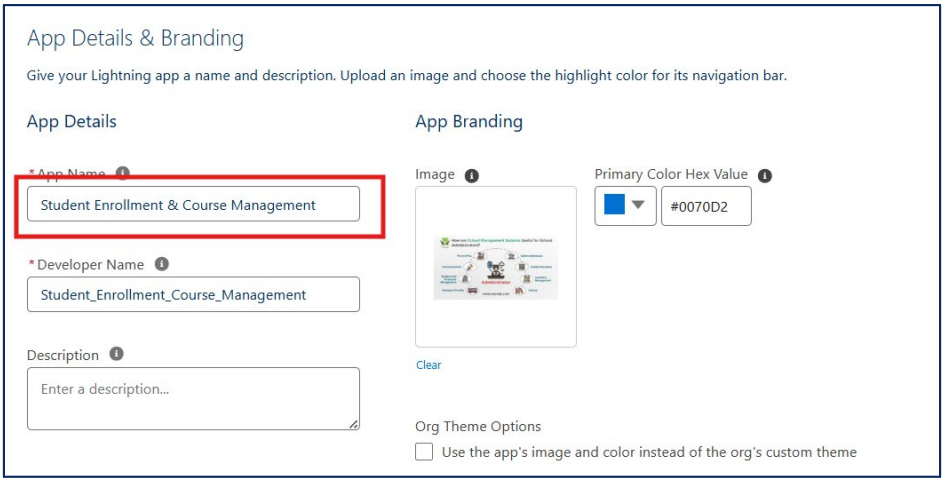


Figure 2. *App Details and Branding in Lightning App Manager: the app name, developer name and primary colour of the SECMS app.*

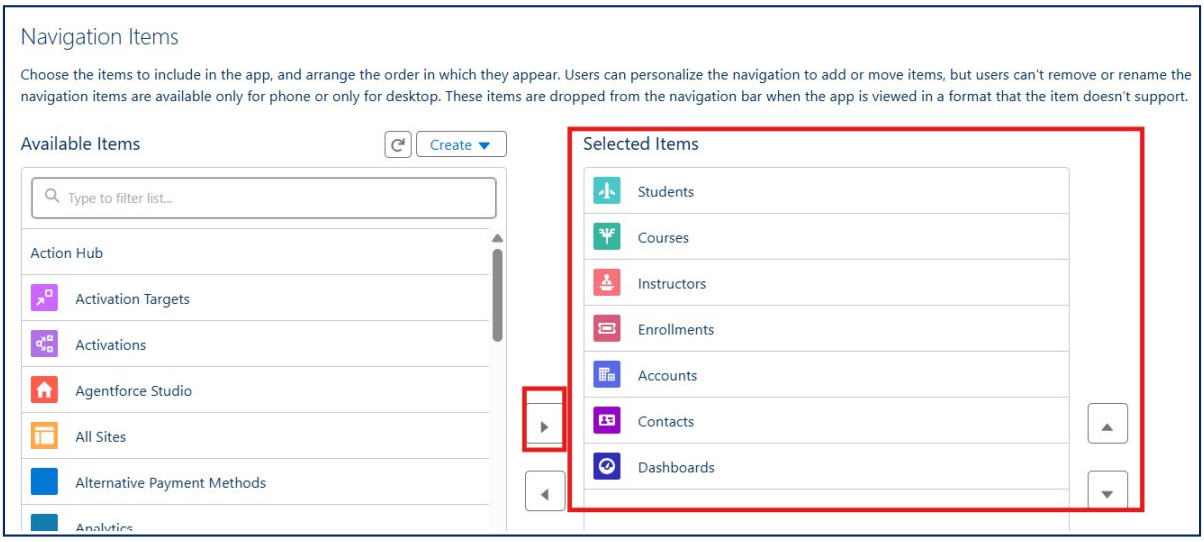


Figure 3. *Navigation Items: Students, Courses, Instructors, Enrollments, Accounts, Contacts and Dashboards are the selected items visible in the screenshot (the list continues beyond the visible area). The order determines the order of tabs in the app.*

3.2 Tabs and Object Pages

Each object has its own tab (Students, Courses, Instructors and Enrollments). Selecting a tab opens the object home page, which shows a list of records and a New button for creating records.

3.3 Record Pages, List Views, Related Lists and Page Layouts

Component	Role in SECMS
Record page	Displays a single record. Its fields and related lists are governed by the page layout (see Section 4).
List view	Shows a filtered set of records from a tab. See Section 8.
Related list	Shows child records on a parent record, for example Enrollments on a Student.
Page layout	Controls field order, sections, required markers, buttons and related lists.

Diagram 1 shows how a user moves through the interface.

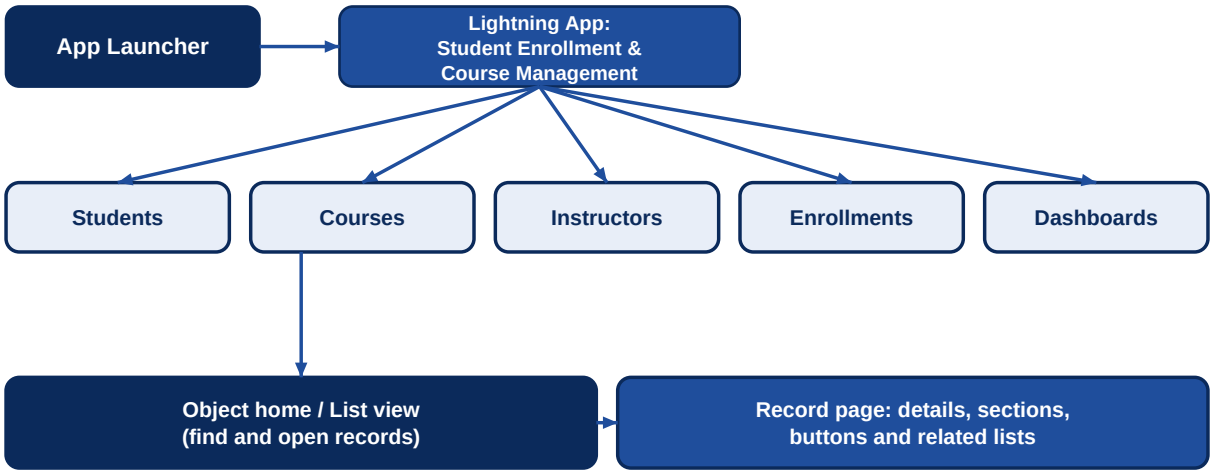


Diagram 1. *Navigation flow from the App Launcher to a record page. The tab list follows the selected navigation items in Figure 3.*

4. Page Layout Customization

Page layouts control which fields, sections, buttons and related lists appear on a record. Enrollment is the only object for which layout screenshots were supplied. Layouts for the other objects are described from the project guide and marked accordingly.

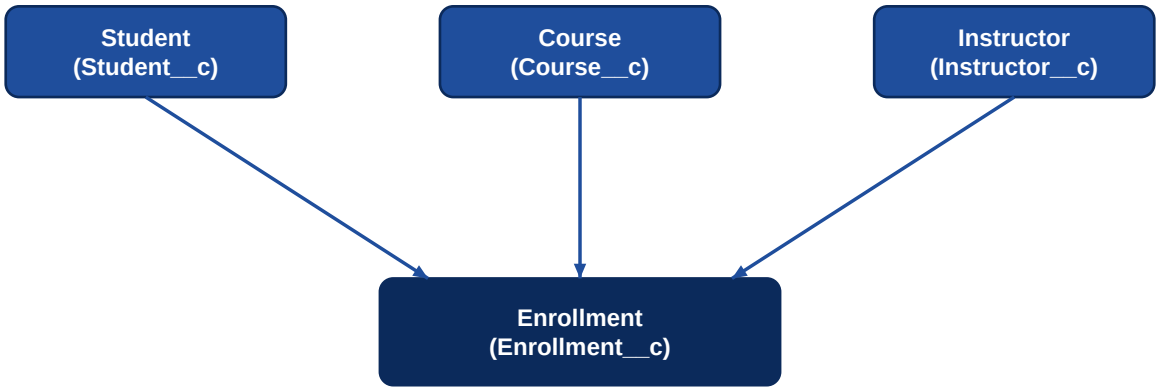


Diagram 2. *Object relationships that drive the layouts and related lists. Each Enrollment stores a Student, a Course and an Instructor through required lookup relationships (Enrollment is the child object).*

4.1 Enrollment (Implemented)

Two record types were created on Enrollment: **New Enrollment** (used for first-time enrollments) and **Re-Enrollment** (used when a student re-enrols in a course). Each has its own page layout, created by cloning the standard Enrollment Layout.

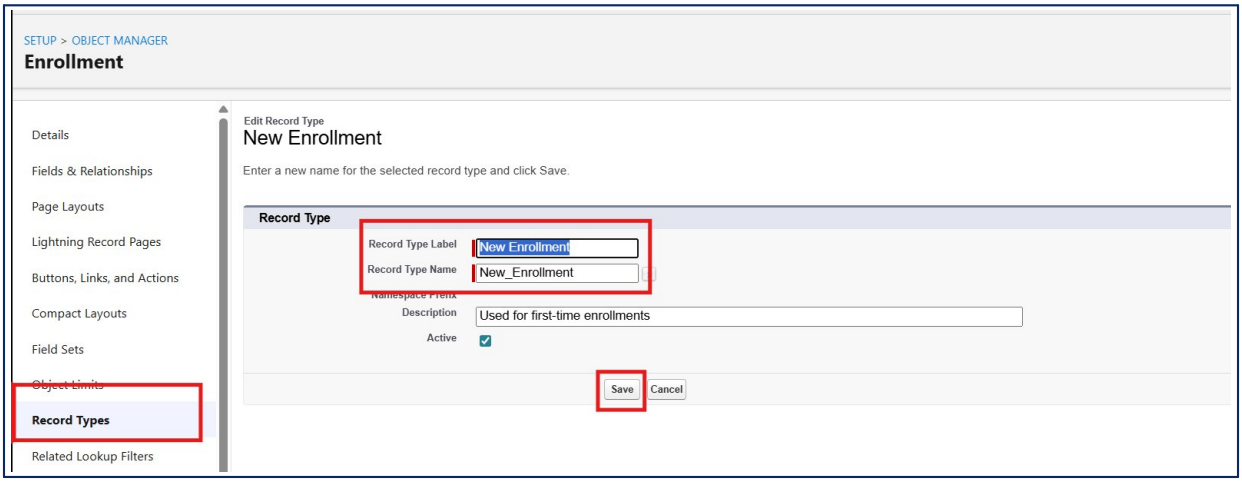


Figure 4. *Record type **New Enrollment** (API name New_Enrollment), described as "Used for first-time enrollments" and set to Active.*

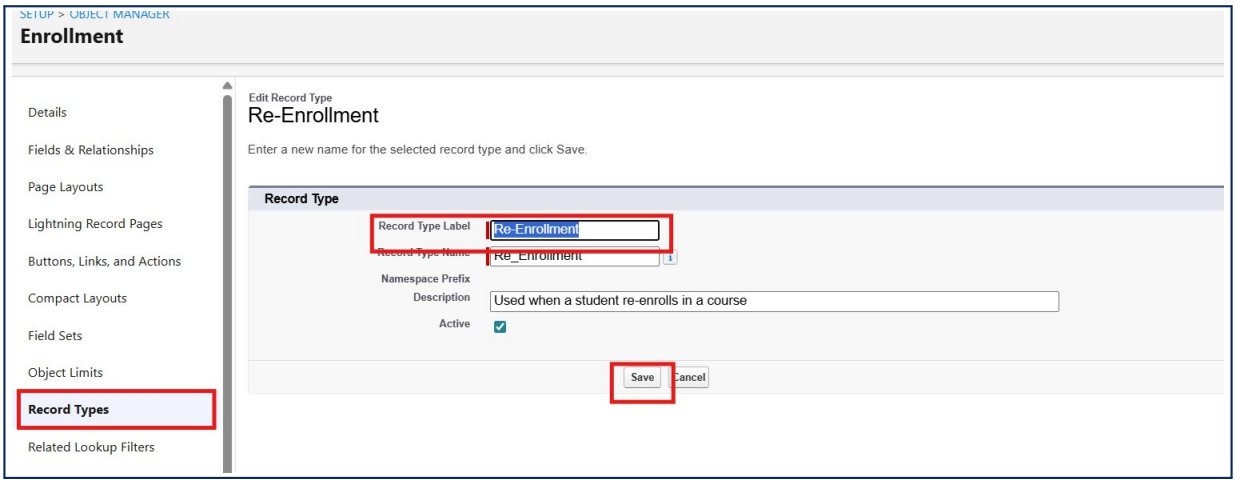


Figure 5. *Record type* ***Re-Enrollment*** *(API name Re_Enrollment), described as "Used when a student re-enrolls in a course" and set to Active.*

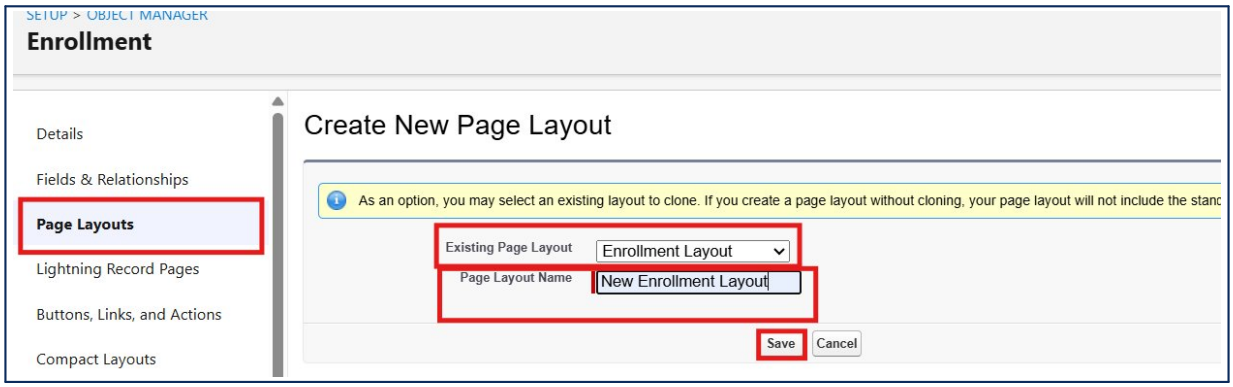


Figure 6. *Creating the New Enrollment Layout by cloning the existing Enrollment Layout.*

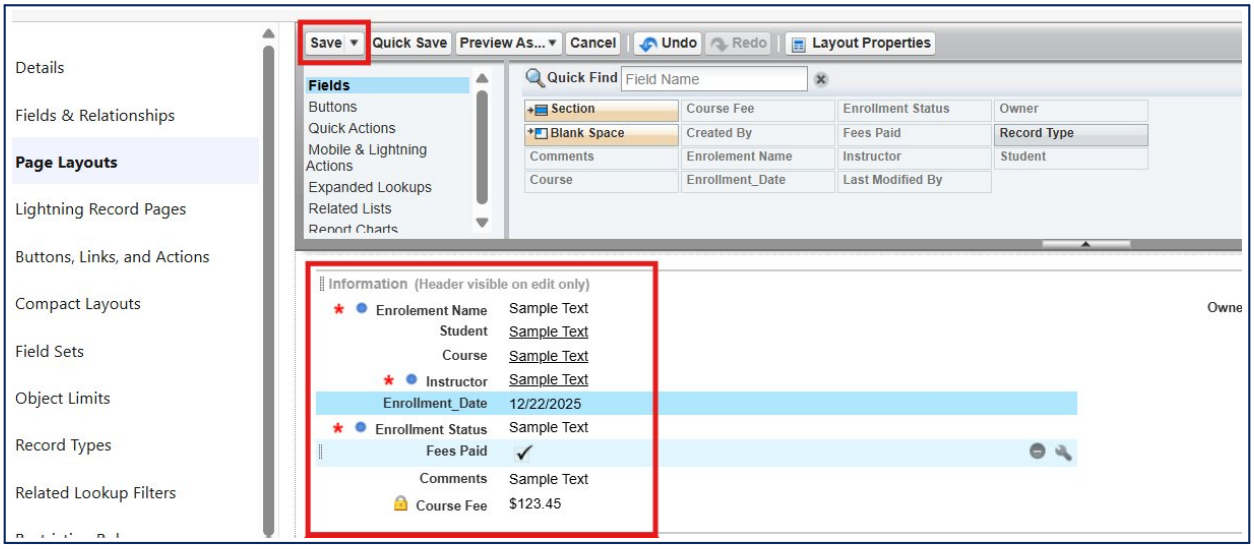


Figure 7. *New Enrollment Layout in the layout editor. The Information section lists Enrolement Name, Student, Course, Instructor, Enrollment_Date, Enrollment Status, Fees Paid, Comments and Course Fee. Required fields carry a red asterisk; Course Fee shows a lock icon (read-only). Owner appears in the right column.*

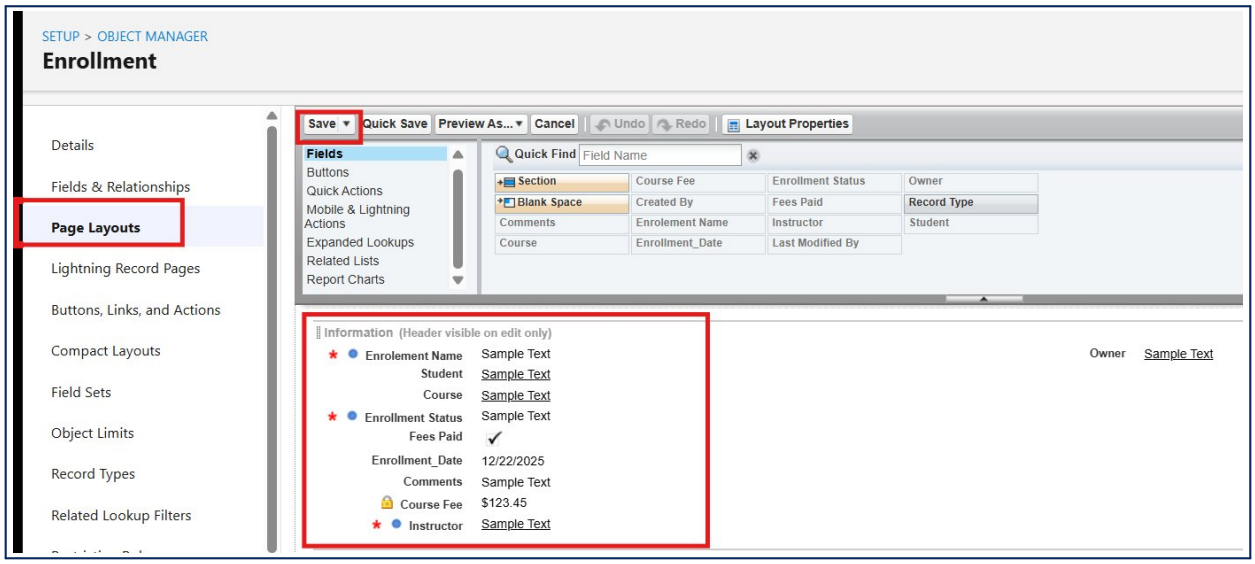


Figure 8. Re-Enrollment Layout in the layout editor. The same fields are used with a different order (Enrollment Status and Fees Paid appear before Enrollment_Date, and Instructor is placed last).

Field arrangement observed in the screenshots:

Position	New Enrollment Layout	Re-Enrollment Layout
1	Enrolement Name (required)	Enrolement Name (required)
2	Student	Student
3	Course	Course
4	Instructor (required)	Enrollment Status (required)
5	Enrollment_Date	Fees Paid
6	Enrollment Status (required)	Enrollment_Date
7	Fees Paid	Comments
8	Comments	Course Fee (read-only)
9	Course Fee (read-only)	Instructor (required)

Requires confirmation. (1) The field is labelled "Enrolement Name" in the layouts (spelling as configured); confirm whether it should be corrected to "Enrollment Name". (2) The project guide mentions a "Re-Enrollment Reason" field and a previous-enrollment reference, but neither appears in the screenshot, so they are not documented as implemented. (3) The layout screenshots show the fields only; the standard buttons and related lists sections were not visible.

4.2 Student, Course and Instructor (Requires confirmation)

The project guide states that the **Enrollments** related list is added to the main page layout of Student, Course and Instructor by dragging it from the Related Lists palette. No screenshots of these layouts were supplied, so the arrangement of their fields and sections cannot be described. The fields defined for these objects in the data model are:

Object	Fields defined in the data model
Student	Student Name, Email, Phone, Date of Birth, Student Status, Address
Course	Course Name, Category, Duration, Course Fee, Description

Object	Fields defined in the data model
Instructor	Instructor Name, Instructor Code, Expertise, Phone, Email
Enrollment	Student, Course, Instructor (lookups), Enrollment Date, Enrollment Status, Fees Paid, Total Amount, Comments

5. Lightning Record Pages

Lightning record pages are configured in Lightning App Builder and control the arrangement of components on a record page (for example Highlights Panel, Details, Related Lists and Tabs). They complement page layouts, which continue to supply the fields and related lists shown by those components.

Status: Planned / requires confirmation. No Lightning App Builder screenshots or activation details were supplied. The Lightning Record Pages entry exists in Object Manager for each object, but no customised page is evidenced. This section therefore describes the general procedure only.

General procedure

- Open the object in Setup > Object Manager and select **Lightning Record Pages**, then click **New**.
- Choose the page type (Record Page) and a template, and enter a label.
- Place components on the canvas, typically a Highlights Panel at the top, a Tabs component with Details and Related tabs, and the Related List component for Enrollments.
- Click **Save**, then **Activation**, and assign the page as the org default, app default or app, record type and profile assignment.
- Open a record to confirm that the new page displays as expected.

Proposed placement for Phase 3 (planned)

Object	Component placement (planned)	Status
Student	Highlights panel; Details tab; Related tab with Enrollments list	Planned
Course	Highlights panel; Details tab; Related tab with Enrollments list	Planned
Instructor	Highlights panel; Details tab; Related tab with Enrollments list	Planned
Enrollment	Highlights panel; Details tab (fields from the assigned layout)	Planned

6. Forms and User Input

Users create and update records through standard Salesforce forms. The form contents follow the page layout assigned to the user's profile (and record type for Enrollment). To create a record, the user opens the relevant tab, clicks **New**, completes the fields, and clicks **Save**. To update, the user opens the record and edits it, then saves.

Form	Fields presented (from the data model)	Notes
Student registration	Student Name, Email, Phone, Date of Birth, Student Status, Address	Email is checked by a validation rule (Figure 10). Layout not shown; requires confirmation.

Form	Fields presented (from the data model)	Notes
Course creation	Course Name, Category, Duration, Course Fee, Description	Layout not shown; requires confirmation.
Enrollment record	Enrolement Name, Student, Course, Instructor, Enrollment_Date, Enrollment Status, Fees Paid, Comments, Course Fee	Record type selects the layout (Figures 7 and 8). Fees Paid rule applies (Figure 11).
Instructor information	Instructor Name, Instructor Code, Expertise, Phone, Email	Layout not shown; requires confirmation.

Automation that affects forms

According to the project guide, a record-triggered flow sets Enrollment_Date to the current date when an enrollment is created, so users need not enter it. The guide also describes flows for instructor assignment, approval emails and follow-up tasks. These belong to the automation phase and are mentioned only because they affect what the user sees on the form.

7. Navigation and User Experience

Users navigate from the App Launcher into the SECMS app and then use the tabs to reach each object. Related lists on a record allow movement to associated records without returning to the tabs.

Usability features (from the evidence)

Improvement	How it is achieved	Evidence
Consistent navigation	One Lightning app; tabs for Students, Courses, Instructors, Enrollments and Dashboards	Figure 3
Clear field labels	Object fields use plain labels such as Student, Course, Instructor, Fees Paid	Figures 7 and 8
Organised page sections	Fields grouped in an Information section on Enrollment layouts	Figures 7 and 8
Required-field markers	Red asterisk on Enrolement Name, Instructor and Enrollment Status	Figures 7 and 8
Helpful validation messages	"Email must be a valid student institution email" and "Fees must be paid before approving an enrollment"	Figures 10 and 11
Separate flows for new and returning students	Record types with dedicated layouts	Figures 4 to 8

8. List Views and Related Lists

8.1 List Views

A list view displays records of one object in a table and can be filtered, sorted and searched. Users open a list view from a tab and can switch between views such as All or Recently Viewed. Each object was set to Allow Search when it was created, which supports finding records from the global search bar.

Requires confirmation. No custom list views were shown in the supplied evidence. Custom views (for example, Pending Enrollments filtered on Enrollment Status = Requested) would be a planned enhancement.

8.2 Related Lists

Because Enrollment holds lookups to Student, Course and Instructor, the Enrollments related list can appear on each of those parent records. The lookup relationship screens show the child relationship name *Enrollments* and the related list label *Enrollments* for Instructor, with the Instructor Layout selected.

Parent record	Related list	What it shows	Status
Student	Enrollments	Courses in which the student is enrolled	Documented
Course	Enrollments	Students enrolled in the course	Documented
Instructor	Enrollments	Enrollments assigned to the instructor	Documented (setup step visible); display not shown

9. User Roles and Access

Page layouts control **what a user sees on a record**; they do not control access. Access is controlled by separate mechanisms:

Mechanism	What it controls
Page layout	Arrangement of fields, sections, buttons and related lists. A field left off a layout is hidden from the page but is not secured by that alone.
Object permissions (profile / permission set)	Whether a user can read, create, edit or delete records of an object.
Field-level security	Whether a user can see or edit an individual field, regardless of layout.
Record access (OWD, sharing, ownership)	Which individual records a user can see.

Interface requirements by role

Role	Interface requirement	Source
Administrator	Access to all objects, tabs, reports and dashboards; maintains layouts and record types	Project guide
Enrollment officer	Create and update enrollments; view students and courses	Project guide
Instructor	Read-only view of assigned enrollments	Project guide

Access settings confirmed by evidence

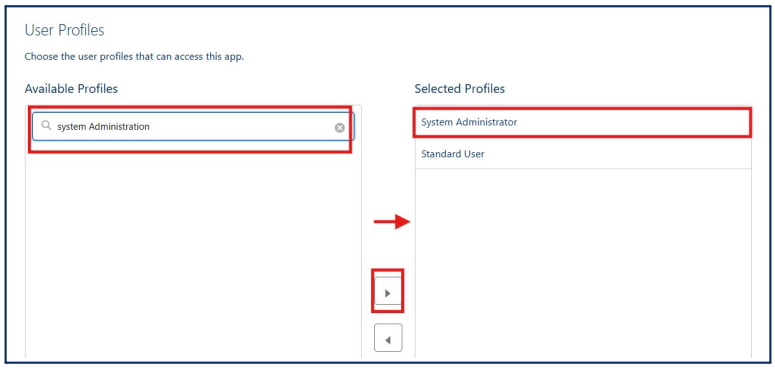


Figure 9. Profiles selected for access to the SECMS Lightning app: System Administrator and Standard User.

- **Confirmed (Figure 9):** the app is assigned to the System Administrator and Standard User profiles.
- **Documented, not evidenced:** the guide lists Enrollment Officer and Instructor profiles, object permissions (Enrollment: Read, Create, Edit for officers), and organisation-wide defaults (Student and Course: Public Read/Write; Enrollment: Private).
- **Requires confirmation:** whether Enrollment Officer and Instructor profiles or permission sets exist in the org, because the app assignment screenshot shows only two profiles. The record types were also said to be enabled for Admin and Enrollment Officer; this is not visible in the supplied screenshots.

10. UI Testing

The test cases below verify the Phase 3 interface. No results have been recorded. The Actual Result column is to be completed during testing, and every Status is currently **Not Executed**.

ID	Test Scenario	Expected Result	Actual Result	Status
UI-01	Open the SECMS app from the App Launcher	App opens and the navigation bar is displayed	To be recorded	Not Executed
UI-02	Open the Students, Courses, Instructors and Enrollments tabs	Each tab opens its object home page without error	To be recorded	Not Executed
UI-03	Open the Dashboards item	Student Management Dashboard is available and loads	To be recorded	Not Executed
UI-04	Click New on Enrollments and select the New Enrollment record type	Form shows the New Enrollment Layout field order	To be recorded	Not Executed
UI-05	Click New on Enrollments and select the Re-Enrollment record type	Form shows the Re-Enrollment Layout field order	To be recorded	Not Executed
UI-06	Check fields on Student, Course and Instructor forms	Fields appear as configured in each layout (to be confirmed)	To be recorded	Not Executed
UI-07	Create a student with valid data and a student-domain email	Record saves and opens the record page	To be recorded	Not Executed
UI-08	Create a student with an email outside the allowed domain	Error message "Email must be a valid student institution email" appears under Email	To be recorded	Not Executed

ID	Test Scenario	Expected Result	Actual Result	Status
UI-09	Create a course with valid data	Record saves successfully	To be recorded	Not Executed
UI-10	Create an instructor with valid data	Record saves successfully	To be recorded	Not Executed
UI-11	Save an enrollment leaving a required field empty (Instructor or Enrollment Status)	Standard required-field message appears and the record is not saved	To be recorded	Not Executed
UI-12	Set Enrollment Status to Approved while Fees Paid is unchecked	Message "Fees must be paid before approving an enrollment" appears at top of page	To be recorded	Not Executed
UI-13	Open a Student record that has enrollments	Enrollments related list shows the student's enrollments (if the list is added)	To be recorded	Not Executed
UI-14	Open a Course and an Instructor record with enrollments	Enrollments related list shows the linked enrollments (if the list is added)	To be recorded	Not Executed
UI-15	Use the record links (Student, Course, Instructor) on an Enrollment	Each link opens the related record	To be recorded	Not Executed
UI-16	Open the app on a desktop browser and on a mobile device or narrow window	Pages remain readable and usable; supported screen sizes to be confirmed	To be recorded	Not Executed

11. Screenshots and Implementation Evidence

Screenshots used in this document come from the project working file. They are listed below with the feature each one supports. Only **after** (configured) views are available. No before-and-after pairs were supplied, and none have been created.

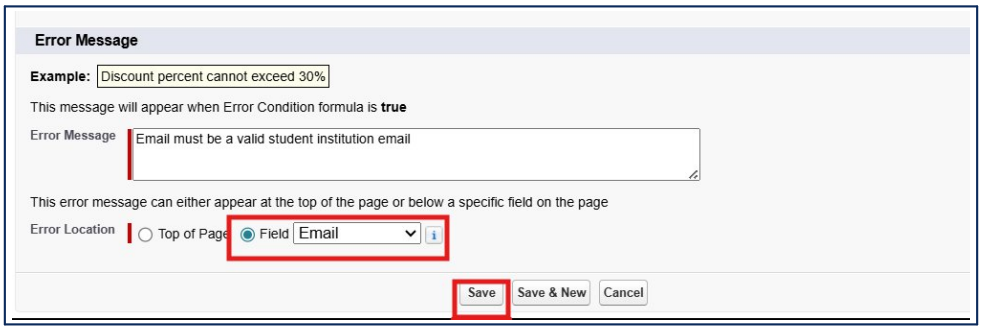


Figure 10. *Validation rule error message for Student email. The message "Email must be a valid student institution email" is shown on the Email field (Error Location: Field).*

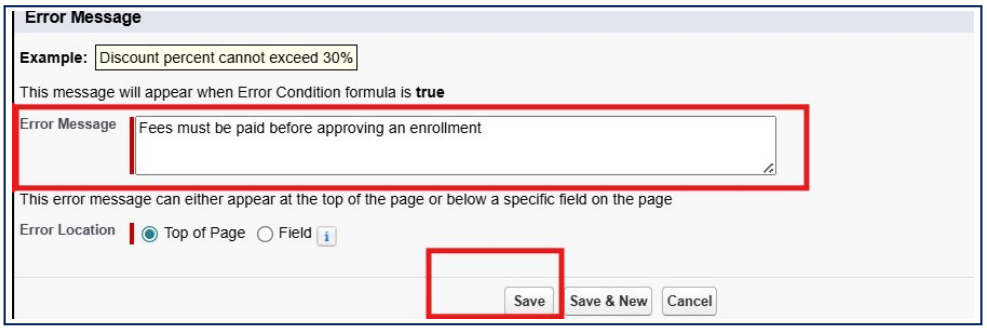


Figure 11. *Validation rule error message for Enrollment. "Fees must be paid before approving an enrollment" is shown at the top of the page (Error Location: Top of Page).*

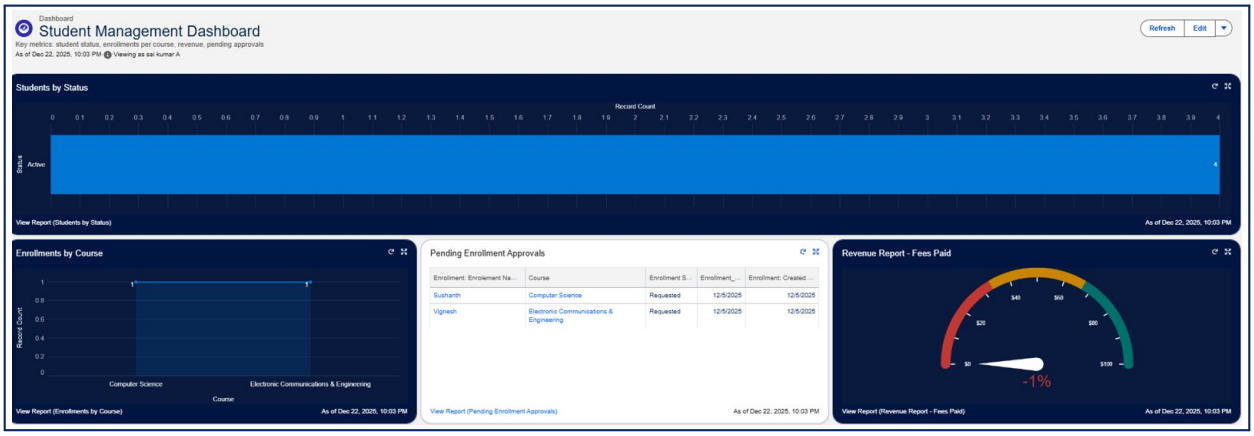


Figure 12. *Student Management Dashboard with Students by Status, Enrollments by Course, Pending Enrollment Approvals (table) and Revenue Report - Fees Paid (gauge). It gives managers a single visual summary.*

Requires confirmation. The rule name shown in the validation screenshot is Email_Must_Be_Student_Domain, while the guide text names it Student_Email_Domain_Validation. The screenshot is treated as the actual configuration. The dashboard gauge displays -1% in the screenshot, which suggests the revenue data needs checking before final submission.

Evidence index

Figure	Feature	Section
1	Students tab definition	3.1
2	Lightning app details and branding	3.1
3	Navigation items	3.1
4	New Enrollment record type	4.1
5	Re-Enrollment record type	4.1
6	Creating the New Enrollment Layout	4.1
7	New Enrollment Layout editor	4.1
8	Re-Enrollment Layout editor	4.1
9	App profile assignment	9
10	Student email validation message	11
11	Fees paid validation message	11
12	Student Management Dashboard	11

12. Phase 3 Deliverables

Deliverable	Status	Remarks
Customized Salesforce page layouts	Partly implemented	Enrollment layouts evidenced; Student, Course, Instructor to be confirmed
Configured Lightning record pages	Planned	No App Builder evidence supplied
Organised fields and related lists	Partly implemented	Enrollment fields organised; related lists documented but not shown
Improved navigation	Implemented	Lightning app and tabs (Figures 1 to 3)
User-friendly record forms	Partly implemented	Enrollment forms and validation messages evidenced
UI testing documentation	Prepared	Test cases written; results pending
Implementation screenshots	Provided	Twelve screenshots from the project file

13. Phase 3 Conclusion

Phase 3 delivered the main interface elements of SECMS: a dedicated Lightning app with tabs for each object, Enrollment record types with separate page layouts, required-field markers, clear validation messages, and a management dashboard. Items that were described but not evidenced, such as layouts for the other objects, related list display, Lightning record pages and role-specific profiles, are marked for confirmation.

These results prepare the application for **Phase 4: Data Migration, Testing and Security**. The interface gives testers a stable set of forms and pages to load data into, the prepared UI test cases can be executed and their results recorded, and the confirmed profile and access questions in Section 9 can be settled when security is configured.

Open items before submission

- Capture screenshots of the Student, Course and Instructor layouts and the Enrollments related lists.
- Decide whether Lightning record pages will be configured in this phase and capture the activation steps.
- Confirm the Enrollment Officer and Instructor profiles and record type enablement.
- Execute UI-01 to UI-16 and record actual results.
- Confirm the spelling "Enrolement Name" on the Enrollment layouts.

References

- Student Enrollment & Course Management System (SECMS) project documentation, supplied working file (Salesforce Developer Edition configuration and screenshots), Team AKASH C.
- No external websites or publications were consulted in preparing this document.